SEASUG Paper 116-2026

Challenges and Opportunities in Creating Modeling-Ready Drug and Device Safety Data from Public Regulatory Sources

Shelly Hunt, Siemens Digital Industry Software

ABSTRACT

Many challenges with publicly available regulatory data from the U.S. Food and Drug Administration (FDA) have been widely noted and documented, although these sources are pivotal to signal detection, pharmacovigilance, post-market surveillance, safety analytics, and risk management. FDA resources including adverse event reports, recalls, warning letters, and clinical trial information are valuable on their own, yet do not provide a complete view of the medical device or drug development lifecycles. These data exist in different repositories, formats, and access mechanisms (API, direct website download, etc.). Once obtained, significant integration barriers remain, including entity resolution challenges and hierarchical data differences.

This work demonstrates how publicly available FDA and related regulatory datasets can be acquired and integrated to create a unified analytics-ready resource. A methodology for retrieving, cleansing, and joining disparate datasets using SLC Workbench is presented. Emphasis is placed on the practical challenges encountered during data preparation, including API limitations, inconsistent product naming conventions, disjointed narrative information, and lack of common join keys. Various entity resolution methods are explored, from traditional matching techniques to emerging LLM-assisted solutions.

INTRODUCTION

Publicly available FDA datasets provide valuable insight into different stages of the regulatory lifecycle. Product approvals, adverse event reports, recall enforcement actions, labeling information, and other regulatory records are available to researchers and the general public. Historically, however, these datasets existed as separate resources distributed across different websites, repositories, reporting systems, and file formats. Researchers attempting to examine a product's lifecycle frequently needed to locate data from multiple sources, learn different access methods, and manually determine how records from independent systems might relate to one another. Even when the data could be obtained successfully, meaningful integration remained difficult because datasets were developed for different regulatory purposes and often lacked common identifiers, consistent structures, or clearly documented relationships.

Since the 2013 Executive Order Making Open and Machine Readable the New Default for Government Information and the subsequent 2019 Open, Public, Electronic, and Necessary (OPEN) Government Data Act, federal agencies have increasingly made public data available in machine-readable formats (Congressional Research Service 2022). The FDA's response to these initiatives was OpenFDA, a centralized platform that provides public access to regulatory datasets through standardized application programming interfaces (APIs). OpenFDA contains information related to approved products, adverse event reports, recalls, medical devices, drug labeling, and numerous other FDA data assets. The continued expansion of OpenFDA creates new opportunities to investigate whether independently maintained regulatory datasets can be combined into broader analytical resources.

The primary objective of this study is to evaluate whether publicly available FDA datasets can be linked to create a unified dataset prepared for analytical exploration of families of drugs or medical devices. Attention is given to dataset integration challenges including identifier availability, entity resolution requirements, hierarchical differences, and inconsistent data structures. Three datasets available through OpenFDA were selected for investigation: the Orange Book, FDA Adverse Event Reporting System (FAERS), and Recall Enforcement Reports. Together, these datasets represent approved products, post-market adverse event reporting, and regulatory action activities.

The central question motivating this work was whether publicly available FDA datasets could be combined to reconstruct a meaningful product lifecycle narrative. Conceptually, a drug progresses through a series of stages that may include initial approval, post-market use, adverse event monitoring,

and potential regulatory intervention such as recalls. Although each of these activities is represented within the FDA's public data ecosystem, they are maintained as independent systems developed for different regulatory purposes. This study therefore evaluates not only whether the datasets can be linked technically, but also whether the resulting relationships are sufficient to support lifecycle-oriented analysis using only publicly available information.

BACKGROUND

PUBLIC FDA DATA SOURCES

The FDA publishes numerous public datasets that document different aspects of the drug and medical device regulatory lifecycle. Although each dataset was developed to support or highlight a specific regulatory purpose, collectively they provide an opportunity to explore how approved products, post-market surveillance activities, and regulatory actions may relate to one another. For this study, three datasets were selected from OpenFDA to represent different points within a proposed product lifecycle framework: Orange Book, FAERS, and Recall Enforcement Reports.

The Orange Book serves as a catalog of approved drug products and provides information on the products, applicant names, active ingredients, application numbers, and current marketing status (U.S. Food and Drug Administration, Orange Book). Because every approved drug ultimately appears within the Orange Book, it was selected as the foundation of this study and served as the starting point for constructing a lifecycle framework. In addition to providing approval information, the dataset contains several product-level identifiers that appeared promising for future cross-dataset integration efforts, including application numbers, active ingredients, brand names, dosage forms, and applicant information.

The FDA Adverse Event Reporting System (FAERS) captures adverse event reports submitted after a product has entered the market (U.S. Food and Drug Administration, FAERS). Unlike the Orange Book, which focuses on approved products, FAERS focuses on reported outcomes and suspected safety concerns associated with product use. Reports may contain multiple drugs, multiple reactions, multiple outcomes, and multiple reporting sources, making the dataset substantially more complex than the product-oriented structure observed within the Orange Book. Because the objective of this study was to explore activities occurring after product approval, FAERS was selected to represent the post-market surveillance portion of the proposed lifecycle framework.

Recall Enforcement Reports provide information regarding FDA regulatory actions, including recall classifications, recall dates, and short form narrative explanations describing why enforcement actions occurred (U.S. Food and Drug Administration, Drug Enforcement Reports). These records are expected to represent the final stage of a straightforward progression from approval to adverse event reporting to recall. The data provide a regulatory perspective not available in either Orange Book or FAERS and are therefore an important component of any lifecycle-oriented analysis.

REGULATORY DATA INTEGRATION CHALLENGES

Although OpenFDA provides a common API framework through which datasets can be retrieved, the datasets themselves were not designed to function as a unified product lifecycle database. Each source was designed for a different regulatory purpose, used different identifiers, and stored information at different levels of granularity. The Orange Book focuses on approved products and regulatory status, FAERS contains every reported adverse event per product, and Recall Enforcement Reports focus on broad-sweeping regulatory actions. As a result, records that appear related from a business perspective are often represented very differently within the underlying data structures.

One of the most significant challenges encountered during this study involved differences in dataset hierarchy. Orange Book is primarily product-centric, where each record represents an approved drug product. In contrast, FAERS is event-centric and allows a single adverse event report to contain multiple drugs, multiple reactions, and multiple outcomes. Recall Enforcement Reports are action-centric and frequently focus on specific manufacturing lots, packaging events, or distribution actions rather than an entire product. Consequently, linking the datasets requires more than simply identifying matching values; it requires understanding how the underlying units of observation differ across sources.

In addition to structural differences, common join keys are largely absent across all three datasets. While several potentially useful identifiers exist, many require normalization, standardization, or additional entity resolution activities before reliable integration can occur. Variables that appear conceptually similar may use different naming conventions, represent different business concepts, or reflect organizational changes that occurred after the original records were created. These challenges motivated the exploratory integration framework developed throughout this study.

EXISTING LITERATURE

Previous research has documented many of the practical challenges associated with FDA regulatory datasets, particularly FAERS (Giunchi et al. 2023). Existing studies frequently focus on pharmacovigilance, signal detection, adverse event monitoring, and safety surveillance activities. While these studies provide valuable insight into downstream analytical applications, considerably less attention has been devoted to the data engineering challenges that must be addressed before those analyses can occur.

A recurring theme throughout the literature is that public regulatory datasets frequently contain structural inconsistencies, duplicate reporting concerns, variable availability differences, and complex hierarchical relationships. These observations align closely with the findings presented in this study. Rather than focusing on statistical modeling or signal detection, this work concentrates on the practical challenges of acquiring, documenting, profiling, and integrating multiple public FDA datasets into a common analytical framework.

The literature also highlights an important gap between data availability and data usability. Public access to regulatory datasets has expanded considerably through initiatives such as OpenFDA, yet relatively little guidance exists regarding how independently maintained datasets can be combined into a coherent analytical resource. Consequently, this study focuses on identifying candidate linkage variables, documenting integration barriers, and evaluating the feasibility of constructing a lifecycle-oriented analytical framework using only publicly available FDA data.

METHODOLOGY

Datasets were selected that could hypothetically provide a comprehensive view of drug approval, post-market surveillance, and regulatory action. Three publicly available FDA data sources were ultimately selected for investigation: Orange Book, FAERS, and Recall Enforcement Reports. Historically these datasets existed in separate systems and were intended to support different regulatory functions. The introduction of OpenFDA in 2015 created an opportunity to access many of these resources through a common API framework. Although OpenFDA was originally developed in response to federal open-data initiatives, new datasets continue to be added and expanded over time. The release of Orange Book data through OpenFDA in June 2026 provided an opportunity to evaluate whether product approval information could connect to adverse events and regulatory enforcement data.

DATA ACQUISITION

Data were retrieved through OpenFDA using SLC Workbench. A workflow was developed using the HTTP procedure to submit HTTPS requests to OpenFDA endpoints and retrieve responses in JSON format. Retrieved responses were subsequently stored and loaded into work library tables through the SLC JSON engine. Initial requests retrieved a sample of one hundred records to support exploratory profiling of dataset structure, metadata, identifiers, and hierarchical relationships.

Example retrieval code:

```
proc http
  url="https://api.fda.gov/drug/orangebook.json?limit=100"
  method="GET"
  out="orange_book_API.json"
  ct="application/json";
run;
```

```
filename orngbk "orange_book_API.json";

libname orngbkjs JSON fileref=orngbk;
```

Similar retrieval workflows were developed for FARES and Recall Enforcement Reports. Data acquisition proved relatively straightforward due to the standardized API framework provided by OpenFDA. Retrieving records, loading JSON responses, and generating relational tables required comparatively little effort. The more significant challenge emerged after acquisition and involved determining how independently developed datasets related to one another and whether meaningful linkage opportunities existed. Consequently, much of the project effort focused on data understanding, profiling, and relationship discovery rather than data retrieval.

DATA PROFILING AND METADATA DOCUMENTATION

Although OpenFDA provides searchable field documentation and endpoint descriptions, understanding how datasets might relate to one another requires extensive exploratory profiling. Candidate linkage variables, identifier patterns, naming conventions, and hierarchical relationships were not always explicitly documented. As a result, much of the investigation relied on examining the observed data itself, identifying recurring patterns, and evaluating whether conceptually similar variables were sufficiently consistent to support cross-dataset integration.

The JSON engine automatically transformed hierarchical JSON responses into collections of relational tables. Following ingestion, metadata were documented through a combination of automated table inspection and FDA-provided searchable field documentation. Variable inventories were created for all business-facing tables generated by the JSON engine. Attention was given to variables that appeared likely to support future integration efforts, including application numbers, active ingredients, brand names, applicant information, manufacturer identifiers, and product attributes.

FAERS required additional profiling because of its hierarchical structure. The dataset contains nested information related to patients, drugs, reactions, outcomes, reporting sources, and event commentary. As a result, a single adverse event report can generate multiple related observations when transformed into a relational structure. Metadata documentation therefore included both business variables and JSON engine generated relationship variables used to preserve parent-child associations across tables. Profiling activities included identification of record-level units of analysis, documentation of parent-child table relationships, review of available identifiers, assessment of variable completeness, and identification of potential cross-dataset linkage candidates. These activities were intended to support the initial screening of potential linkage variables as described in the following sections.

PRELIMINARY INTEGRATION ASSESSMENT

Candidate Matching Variable Framework

During profiling activities, a framework was developed to evaluate variables that appear capable of linking records across datasets. Rather than focusing on individual variable names, candidate variables were grouped according to common business concepts. For example, the Orange Book variable `application_number`, FAERS variable `drugauthorizationnumb`, and Recall Enforcement Reports variable `application_number1` all represent the same underlying regulatory concept despite differences in naming conventions and formatting. Organizing variables by business concept allowed potential linkage opportunities to be evaluated consistently across datasets.

Candidate variables were ultimately divided into two broad categories. Human-readable identifiers included active ingredients, brand names, and manufacturer information. These variables can generally be interpreted without specialized regulatory knowledge and often provide meaningful business context through manual inspection. FDA-assigned identifiers included application number, product NDC, package NDC, RXCUI, SPL ID, SPL set ID, and UNII. Unlike the human-readable identifiers, these variables primarily serve regulatory, interoperability, and product-identification purposes and typically require supporting documentation or domain knowledge to interpret. Together, these identifiers formed the basis of the preliminary integration assessment and were later evaluated through the EDARBI case study.

Confidence Assessment Methodology

Each candidate linkage variable was assigned a qualitative confidence level based on manual review of available records. Variables were evaluated according to four criteria:

- Availability across datasets
- Consistency of observed values
- Required transformation effort
- Entity resolution requirements

Confidence ratings of Very High, High, Medium, and Low were used to summarize preliminary linkage potential. These ratings were intended as practical assessments derived from exploratory profiling and record inspection activities rather than formal statistical measures.

To support the integration assessment, observed matches were documented within a linkage evidence worksheet. Example values were recorded across datasets along with required transformations, observed inconsistencies, and potential integration barriers. This process provided traceability for linkage decisions and facilitated later evaluation of candidate variables.

CASE STUDY INVESTIGATION

A focused case study was conducted using the drug product EDARBI (Azilsartan). The objective of the case study was to evaluate whether candidate linkage variables identified during profiling could successfully connect records across Orange Book, FAERS, and Recall Enforcement Reports. Application number was selected as the primary retrieval variable because it demonstrated strong availability across all three datasets during profiling activities. Known EDARBI application numbers were subsequently used to retrieve product, adverse event, and recall records from their respective data sources. Dataset retrieval was performed using OpenFDA API queries constructed around the identified application number.

Example EDARBI retrieval code:

```
proc http

url="https://api.fda.gov/drug/event.json?search=patient.drug.drugauthorizat
ionnumb:200796&limit=100"
    method="GET"
    out="faers_API.json"
    ct="application/json";
run;

filename faers "faers_API.json";
libname faersjs JSON fileref=faers;
```

The previous code pulls 100 records in each the Orange Book, FAERS, and Recall Enforcement Reports for EDARBI utilizing the known FDA application number 200796. The JSON engine then assigned each relational table to their respective generated libraries, `faersjs`, `orngbkjs`, and `recalljs`.

Human-Readable Identifier Evaluation

Following retrieval, candidate linkage variables were evaluated through a combination of visual inspection and exploratory queries. Emphasis was placed on active ingredient, brand name, manufacturer information, and application number because these variables emerged as promising linkage candidates during profiling. These variables appeared most generally interpretable to both subject matter experts and public users. Exploration activities included review of raw records, identification of recurring values, evaluation of naming consistency, and comparison of candidate linkage variables across datasets. Additional queries were developed to examine distinct values and support manual validation efforts. These activities were intended to assess whether conceptually similar variables were sufficiently consistent to support future integration efforts.

Sample active ingredient investigation code:

```
proc sql;
create table ob_active as
select distinct upcase(name) as active_ingredient
from orngbkjs.products_active_ingredients
order by active_ingredient;
quit;

proc sql;
create table faers_active as
select distinct upcase(activesubstancename) as active_ingredient
from faersjs.drug_activesubstance
order by active_ingredient;
quit;

proc sql;
create table recall_active as
select distinct upcase(substance_name1) as active_ingredient
from recalljs.openfda_substance_name
order by active_ingredient;
quit;
```

RESULTS

DATA STRUCTURE AND PROFILING RESULTS

Data acquisition and ingestion were successful across all three datasets. OpenFDA provided a relatively straightforward mechanism for retrieving records through HTTPS requests, regardless of whether the underlying source was originally made available through bulk downloads, interactive web reports, or other distribution methods. The technical challenge was not obtaining the data itself. Instead, the more significant challenge involved understanding how independently developed datasets were structured, determining how they related to one another, and identifying variables that might support integration efforts.

One observation that became apparent early in the profiling process was the limited amount of comprehensive documentation available for some portions of the datasets. Although OpenFDA provides searchable field definitions and endpoint documentation, users are often required to navigate back to the originating FDA datasets to obtain additional context. Variable descriptions are generally available for searchable fields; however, understanding the meaning and purpose of many additional variables requires substantial independent investigation and exploratory profiling. As a result, data understanding became a major component of the overall effort.

Orange Book Findings

The Orange Book API retrieval generated multiple hierarchically related tables that were subsequently transformed into SLC library tables through the JSON engine. In addition to business-facing tables, the JSON engine automatically created helper tables such as `ROOT`, `META`, and `META_RESULTS`, including variables with the prefix `ORDINAL_`. These structures were not part of the original FDA data but instead provided relationship information used to navigate the hierarchical source document. Business-facing Orange Book data were ultimately distributed across tables containing product information, active ingredients, therapeutic equivalence information, patents, and approval-related metadata.

A major preliminary finding is that the `PRODUCTS_ACTIVE_INGREDIENTS` table is remarkably easy to interpret. The table contains active ingredient names and strengths, such as “HYDROCHLOROTHIAZIDE 12.5MG” or “LEVODOPA 250MG”. These variables are highly interpretable and require little domain expertise to understand. In contrast, the `PRODUCTS_THERAPEUTIC_EQUIVALENCE` table contained abbreviated classification codes such as “AB3” and “AP”. Although clearly important from a regulatory

perspective, the meaning of these codes was not immediately apparent from the data itself and required additional reference documentation to interpret. Similar patterns appeared elsewhere in the dataset. Several tables contained approval dates, product numbers, and regulatory metadata, but understanding the business meaning of those values often required navigation across related tables. Individual variables were frequently meaningful only within the context of associated product records. This reinforced the importance of understanding the parent-child relationships generated during JSON ingestion.

The `RESULTS_PRODUCTS` table quickly emerged as the most valuable table for integration purposes. It contained many of the business variables that originally motivated selection of the Orange Book dataset, including application number, brand name, application name, dosage form, applicant information, and marketing status. Particularly noteworthy was the ability to identify products with statuses such as “DISCONTINUED”. Because these variables provide both product-level context and potential linkage candidates, the table served as a central reference point throughout the remainder of the investigation.

Perhaps the most unexpected finding was the overall simplicity of the Orange Book dataset. Given its role as the authoritative catalog of FDA-approved drug products, a much more complex structure was initially anticipated. Instead, the dataset was largely focused on product identity, approval information, and regulatory status. This simplicity ultimately proved advantageous because it provided a stable product-centric reference layer from which more complicated datasets, such as FAERS and Recall Enforcement Reports, could be explored.

Adverse Event Findings

Unlike the Orange Book, which was relatively concise and product-oriented, FAERS immediately revealed a much more complex and highly nested structure. The dataset contains information related to patients, drugs, reactions, outcomes, reporting sources, and additional event-level metadata. As a result, a single adverse event report can expand into multiple related observations when transformed from JSON into a relational format.

Adverse event reports should also be interpreted within the context of the FDA's reporting system. FAERS records represent submitted reports of adverse events and observed outcomes rather than verified causal relationships. The presence of a report does not establish that a particular drug caused a particular outcome, nor should the data be used to support medical decision-making. For the purposes of this study, FAERS records were evaluated as regulatory data artifacts that may contribute to lifecycle analysis rather than as evidence of product effectiveness or safety.

One of the most notable findings during profiling involved the OpenFDA harmonization fields. These fields contain collections of related reference information such as brand names, manufacturer names, generic names, product NDCs, package NDCs, and other identifiers. Unlike traditional relational tables, these fields often contain variable-length arrays that can differ dramatically from one record to the next. During JSON ingestion, the SLC JSON engine flattened these arrays into numbered variables. For example, a repeating array containing brand names may be transformed into variables `BRAND_NAME1` through `BRAND_NAME323`, while a manufacturer array may generate variables `MANUFACTURER_NAME1` through `MANUFACTURER_NAME434`. Rather than representing individual observations, these variables represent repeated values originating from a single source record. Consequently, a relatively small number of records can generate an unexpectedly large number of columns within the resulting relational tables.

A useful example of this phenomenon can be observed in the OpenFDA brand name array. One acetaminophen record contained 323 distinct brand names. When transformed into a relational structure, the JSON engine created 323 separate brand name variables rather than 323 observations. The resulting fields contained well-known product names such as Tylenol, Panadol, Feverall, and numerous other retailer-specific private-label products. This single source record therefore generated hundreds of columns within the resulting table structure. Table 1 summarizes the distribution of brand name array sizes observed during profiling of the random 100 observation record API pull.

Statistic	Value
N	208
Minimum	1
25th Percentile	1
Median	1
Mean	11.46
75th Percentile	2
Maximum	323

Table 1. Distribution of Brand Name Array Sizes

Although the brand name arrays were informative, the manufacturer name arrays proved even more interesting. Manufacturer information exhibited similar behavior but expanded to an even greater extent. While most records contained relatively few manufacturers, several records contained hundreds of associated organizations. These included manufacturers, distributors, suppliers, healthcare organizations, and other entities linked through OpenFDA harmonization processes. Table 2 summarizes the distribution of manufacturer name array sizes.

Statistic	Value
N	208
Minimum	1
25th Percentile	1
Median	1
Mean	18.48
75th Percentile	10
Maximum	434

Table 2. Distribution of Manufacturer Name Array Sizes

Figure 1 illustrates a portion of the largest manufacturer-name array identified during profiling and demonstrates how a single JSON array can expand into hundreds of variables when converted to a relational structure.

9/21/26, 3:26 PM Altair SLC ODS MARKUP Output

Altair SLC

Obs	_NAME_	COL1	COL2
1	manufacturer_name1	Apria Healthcare LLC	Apria Healthcare LLC
2	manufacturer_name10	Superior Oxygen & Medical Equipment	Superior Oxygen & Medical Equipment
3	manufacturer_name100	Garrett County Memorial Hospital	Garrett County Memorial Hospital
4	manufacturer_name101	Surgimed Corporation	Surgimed Corporation
5	manufacturer_name102	Olympic Ambulance Inc. DBA Olympic Oxygen Service	Olympic Ambulance Inc. DBA Olympic Oxygen Service
6	manufacturer_name103	Med-Caire Inc.	Med-Caire Inc.
7	manufacturer_name104	Atlas Welding Supply Company, Inc.	Atlas Welding Supply Company, Inc.
8	manufacturer_name105	Timberlake Respiratory Care and Home Medical	Timberlake Respiratory Care and Home Medical
9	manufacturer_name106	MasVida Health Care Solutions, LLC	MasVida Health Care Solutions, LLC
10	manufacturer_name107	JW Welding Supplies & Tools	JW Welding Supplies & Tools
11	manufacturer_name108	WEST PENN LACO, INC.	WEST PENN LACO, INC.
12	manufacturer_name109	IWS Gas and Supply of Texas, LTD	IWS Gas and Supply of Texas, LTD
13	manufacturer_name11	Indiana Oxygen Company Inc	Indiana Oxygen Company Inc
14	manufacturer_name110	Alpha Oxy-Med LLC	Alpha Oxy-Med LLC
15	manufacturer_name111	Whole Gas Technologies LLC	Whole Gas Technologies LLC
16	manufacturer_name112	BayCare Home Care, Inc.	BayCare Home Care, Inc.
17	manufacturer_name113	Air Liquide Canada Inc	Air Liquide Canada Inc
18	manufacturer_name114	B & J Welding Supply, LTD. dba B & J Welding Supply	B & J Welding Supply, LTD. dba B & J Welding Supply
19	manufacturer_name115	Express Medical Inc	Express Medical Inc
20	manufacturer_name116	Mountain Aire Medical Supply, Inc	Mountain Aire Medical Supply, Inc

Figure 1. Example of Manufacturer Name Array Expansion

Examination of the `OPENFDA_MANUFACTURER_NAME` table revealed that although the median record contained only a single manufacturer and 75% of records contained ten or fewer manufacturers, one record contained 434 distinct manufacturer names. Inspection of this record revealed hundreds of oxygen suppliers, medical gas distributors, respiratory care organizations, and home healthcare providers. When flattened into a relational format, these values generated variables `MANUFACTURER_NAME1` through `MANUFACTURER_NAME434`. This finding further illustrates how a relatively small number of atypically large arrays can substantially increase the dimensionality of relational datasets generated from semi-structured JSON sources. To better understand the scope of dimensional expansion occurring throughout the OpenFDA harmonization fields, several commonly populated arrays were profiled and compared. Results are presented in Table 3.

Array Type	N	Minimum	25th Percentile	Median	Mean	75th Percentile	Maximum
Brand Name	208	1	1	1	11.46	2	323
Manufacturer Name	208	1	1	1	18.48	10	434
Package NDC	208	1	10	10	132.91	49	4,054
Generic Name	207	1	1	1	1.33	1	10
Application Number	207	1	1	1	6.47	6	71
NUI	157	1	2	2	2.41	2	7

Table 3. Distribution of OpenFDA Array Sizes After JSON Flattening

Several patterns become apparent. Generic name and NUI arrays generally remained small, with median values of one and two populated elements, respectively. Application number arrays were moderately larger, exhibiting a maximum size of seventy-one values. In contrast, brand name, manufacturer name, and package NDC arrays demonstrated substantially greater variability and dimensional expansion.

The most extreme example involved the package NDC array. National Drug Codes (NDCs) are FDA-recognized product identifiers used to distinguish drug products and packaging configurations. Product NDCs identify the product itself, while package NDCs provide additional granularity by identifying specific package presentations. Although the median record contained only ten package NDC values, one record contained 4,054 populated entries. When transformed into a relational table, this single record generated over four thousand package identifier variables. The mean package NDC array size of 132.9, compared with a median of only 10, further illustrates the highly skewed nature of the data and indicates that a relatively small number of records are responsible for much of the observed dimensional expansion.

Collectively, these findings highlight one of the primary challenges associated with semi-structured FDA data sources. While many records contain only a small number of repeated values, a small subset contains extremely large arrays capable of generating hundreds or even thousands of variables during relational conversion. Such characteristics have important implications for profiling, storage, integration, and downstream analytical efforts.

Another unexpected finding emerged during the EDARBI investigation. Filtering FAERS records for a specific product did not isolate the target drug alone. Because adverse event reports may contain multiple medications taken by the same patient, many records contained both EDARBI and additional co-administered products. As a result, subordinate drug tables frequently included medications unrelated to the product under investigation. The implications of this finding become more apparent during the EDARBI case study, discussed later in this paper.

These observations reinforce one of the major themes that emerged throughout the study: FAERS is fundamentally event-centric rather than product-centric. While Orange Book focuses on approved products, FAERS focuses on reported events, and each event may reference multiple drugs, reactions, and outcomes simultaneously. This difference in analytical granularity would later prove to be one of the most significant integration challenges identified during the study.

Recall Enforcement Report Findings

Although Orange Book provides information regarding product approval and current status, Recall Enforcement Reports provide something that was largely absent from the other datasets investigated: narrative context. In particular, the `reason_for_recall` variable contains descriptive explanations documenting why regulatory action occurred. This information provided a level of detail that was not readily available within either Orange Book or the OpenFDA adverse event endpoint utilized during this study. Adverse event reports typically describe what was reported, whereas Recall Enforcement Reports often provide greater insight into why corrective action was initiated.

Recall Enforcement Reports are fundamentally centered around regulatory actions. Each record represents a specific recall event and contains information such as recall classification, enforcement status, initiation dates, termination dates, manufacturer details, and narrative descriptions. In contrast, each FAERS record represents an individual adverse event report. This distinction initially appeared straightforward; however, profiling activities revealed that the relationship between adverse events and recall actions was considerably more complex than originally anticipated. One of the first observations was that Recall Enforcement Reports contain a substantial number of FDA-assigned identifiers, including NDCs, RXCUIs, UNIIs, recall numbers, and event IDs. Compared with Orange Book, the dataset was generally less intuitive to interpret through manual review alone. At the same time, many of these identifiers were also present in other datasets investigated throughout the study. This immediately suggested that although the data may be less human-readable, it could prove highly valuable for integration purposes.

Another important finding involved the amount of lifecycle information contained within recall records. In addition to recall classifications such as “Class I”, “Class II”, or “Class III”, records include recall status information, initiation dates, termination dates, and enforcement details. Most importantly, Recall Enforcement Reports contain narrative descriptions explaining the reason for the recall. These narratives provide context regarding packaging defects, labeling issues, contamination events, sterility concerns, manufacturing deviations, and other regulatory findings. Whereas Orange Book may indicate that a product was recalled, Recall Enforcement Reports provide insight into the circumstances that led to that action.

The narrative fields were among the most informative variables identified during the investigation, but they also introduced new challenges. Differences in terminology, variable levels of detail, inconsistent naming conventions, and lengthy free-text descriptions make these fields considerably more difficult to standardize than structured identifiers. Consequently, narrative information is highly valuable from an analytical perspective while simultaneously presenting potential challenges for future integration efforts.

One of the most notable findings of the entire study emerged during profiling of Recall Enforcement Reports. The original lifecycle hypothesis assumed that adverse event reports and recall actions would represent sequential stages of a common product narrative. However, examination of recall records suggested that many regulatory actions were not driven by adverse event reporting trends. Instead, recalls frequently originated from manufacturing issues, packaging defects, labeling discrepancies, contamination concerns, quality-control problems, or inspection findings. This observation suggests that recall activity often reflects operational and regulatory processes that are separate from patient-reported outcomes.

Initial assumptions also suggested that recalls would primarily represent product-level outcomes. Profiling revealed a more complicated reality. Many recall records were associated with specific manufacturing lots, packaging activities, distribution actions, or limited subsets of affected products rather than an entire drug product. As a result, recall activity frequently occurred at a different level of analysis than originally anticipated.

This finding ultimately challenged the original lifecycle model that motivated the study. Rather than representing a simple progression from product approval to adverse event reporting and finally to recall, the data suggest that regulatory enforcement activities may follow pathways that are partially independent of adverse event reporting systems. While relationships between adverse event reports and recalls certainly exist, those relationships appear less direct and less deterministic than originally hypothesized.

CANDIDATE LINKAGE VARIABLE ASSESSMENT

Before evaluating individual linkage variables, an important observation emerged during profiling activities. The Orange Book, FAERS, and Recall Enforcement Reports do not simply store different information about the same entities; they operate at fundamentally different levels of analysis. Orange Book is product-centric, FAERS is event-centric, and Recall Enforcement Reports are action-centric. Consequently, many of the integration challenges observed later in the study were driven not by identifier quality but by differences in the underlying units of observation. This finding became one of the most important considerations when evaluating candidate linkage variables.

Human-Readable Identifiers

One objective of the integration assessment was to identify variables that could be readily understood by subject matter experts, analysts, and members of the general public. These variables included drug names, active ingredients, and manufacturer information. Unlike FDA-assigned identifiers, human-readable variables are immediately interpretable and can often be manually verified without reference documentation. However, these variables are also more vulnerable to spelling differences, naming variation, organizational changes, and other entity resolution challenges. As a result, their usefulness depends heavily on consistency and standardization across data sources.

Active ingredients and brand names demonstrated stronger linkage potential than originally anticipated. In contrast, manufacturer information quickly exposed organizational complexities that were not immediately apparent during initial profiling activities. Table 4 summarizes the human-readable identifiers evaluated during the preliminary integration assessment along with variable availability, transformation requirements, and preliminary linkage assessments.

Active Ingredient	
Orange Book Variable	`name`
FAERS Variable	`activesubstancename`
Recall Variable	`substance_name1`
Availability Across Sources	Yes
Exact Match Observed	Yes
Transformation Required	Minimal
Entity Resolution Required	No
Preliminary Assessment	Very High
Brand Name	
Orange Book Variable	`brand_name`
FAERS Variable	`medicinalproduct`
Recall Variable	`brand_name1`
Availability Across Sources	Yes
Exact Match Observed	Yes
Transformation Required	Potentially
Entity Resolution Required	Potentially
Preliminary Assessment	Very High
Manufacturer	
Orange Book Variable	`application_full_name`
FAERS Variable	`companynumb`
Recall Variable	`manufacturer_name1`
Availability Across Sources	Partial
Exact Match Observed	No
Transformation Required	Yes
Entity Resolution Required	Yes
Preliminary Assessment	Medium

Table 4. Human-Readable Identifier Assessment Results

Human-readable variables generally performed better than originally anticipated. Active ingredients and brand names all demonstrated repeatable cross-dataset linkage opportunities, requiring little transformation and minimal manual intervention. Manufacturer information proved to be the primary exception and exposed several entity-resolution challenges discussed later in the EDARBI case study.

Active ingredients are also available in all three datasets. These are generally human interpretable and represent chemical identity rather than marketing identity. These names are less sensitive to branding changes. As explored in the Orange Book dataset, one generic drug may have hundreds of commonly known brand names, but the scientifically standard generic name will rarely change or have many differences in spelling/formatting. As a result, active ingredients are more stable than product names or

manufacturer names. Brand names are also available across all datasets and are highly intuitive to subject matter experts or the common person. Major challenges in addition to multiple brand names for the same ingredients include product rebranding, spelling differences, and abbreviations.

Overall, human-readable variables performed significantly better than initially expected. Active ingredients and brand names all demonstrated successful cross-dataset linkage during the EDARBI investigation. These variables required little or no transformation and were easily verified through manual inspection. Manufacturer information represented the primary exception and quickly exposed many of the entity resolution challenges associated with real-world regulatory data integration. Although human-readable variables may not always provide deterministic linkage opportunities, they remain valuable because they are intuitive, interpretable, and widely available across multiple FDA datasets.

FDA-Assigned Identifiers

Although FDA-assigned identifiers are generally less intuitive than the business-facing fields, they are specifically designed to support standardization and product identification. Variables such as product NDC, package NDC, RXCUI, SPL ID, and SPL set ID initially appeared difficult to interpret. However, exploratory analysis suggested that these identifiers may ultimately provide more reliable linkage opportunities than many human-readable variables. Unlike manufacturer names or brand names, which may change over time due to business decisions, acquisitions, or marketing activities, FDA-assigned identifiers are designed to remain stable and uniquely identify products or product concepts. Consequently, these identifiers were evaluated as potential alternatives to traditional business-oriented matching approaches. Table 5 summarizes the FDA-assigned identifiers evaluated during the preliminary integration assessment and documents observed availability, exact matches, and preliminary linkage assessments.

Application Number	
Orange Book Variable	`application_number`
FAERS Variable	`drugauthorizationnumb`
Recall Variable	`application_number1`
Availability Across Sources	Yes
Exact Match Observed	Yes
Transformation Required	Minor (e.g., NDA200796 vs 200796)
Entity Resolution Required	No
Preliminary Assessment	Very High
Product NDC	
Orange Book Variable	N/A
FAERS Variable	`product_ndc1`
Recall Variable	`product_ndc1`
Availability Across Sources	FAERS, Recall
Exact Match Observed	Yes
Transformation Required	No
Entity Resolution Required	No
Preliminary Assessment	Very High
Package NDC	
Orange Book Variable	N/A

FAERS Variable	package_ndc1
Recall Variable	package_ndc1
Availability Across Sources	FAERS, Recall
Exact Match Observed	Yes
Transformation Required	No
Entity Resolution Required	No
Preliminary Assessment	Very High
RXCUI	
Orange Book Variable	N/A
FAERS Variable	rxcui1
Recall Variable	rxcui1
Availability Across Sources	FAERS, Recall
Exact Match Observed	Yes
Transformation Required	No
Entity Resolution Required	No
Preliminary Assessment	Very High
SPL ID	
Orange Book Variable	N/A
FAERS Variable	spl_id1
Recall Variable	spl_id1
Availability Across Sources	FAERS, Recall
Exact Match Observed	Yes
Transformation Required	No
Entity Resolution Required	No
Preliminary Assessment	Very High
SPL Set ID	
Orange Book Variable	N/A
FAERS Variable	spl_set_id1
Recall Variable	spl_set_id1
Availability Across Sources	FAERS, Recall
Exact Match Observed	Yes
Transformation Required	No
Entity Resolution Required	No
Preliminary Assessment	Very High
UNII	

Orange Book Variable	N/A
FAERS Variable	`unii1`
Recall Variable	N/A
Availability Across Sources	Limited
Exact Match Observed	Unknown
Transformation Required	N/A
Entity Resolution Required	No
Preliminary Assessment	Medium

Table 5. FDA-Assigned Identifier Assessment Results

Application numbers were successfully identified across all datasets: Orange Book, FAERS, and Recall. There are some minimum formatting differences, for example recall datasets prepend application type prefixes to application numbers such as “NDA” and “ANDA.” Although these are minimum formatting differences, it’s not a true source of major difference. Simple normalization may resolve the issue. This variable appears to be the strongest linkage candidates discovered. Application numbers represent FDA-assigned regulatory identifiers that remain sufficiently concise and interpretable to support manual validation efforts.

Product NDC and package NDC demonstrated some of the strongest linkage potential observed during the investigation. In a few instances, exact overlap was observed between FAERS and Recall data. Little additional processing is required beyond normal profiling activities. Product NDC identifiers appeared useful for product-level linkage, while package NDC identifiers provide additional granularity by identifying specific packaging configurations.

RXCUI values also performed well during evaluation. RXCUIs are standardized RxNorm concept identifiers used to represent drugs consistently across healthcare systems and support interoperability among drug information systems (National Library of Medicine 2024). Although these identifiers are less intuitive than the variables such as brand name or active ingredient, exact overlaps were observed between datasets. Because RXCUI values are derived from a standardized vocabulary intended to support drug interoperability, they appear particularly promising for automated integration workflows.

The strongest FDA-assigned identifiers observed during the investigation were SPL ID and SPL set ID. SPL (Structured Product Labeling) is the FDA's standardized electronic format for exchanging product labeling information (U.S. Food and Drug Administration, Structured Product Labeling Resources). An SPL ID uniquely identifies a specific labeling document, while an SPL set ID identifies the broader labeling record and remains stable across revisions to that document. Both identifiers demonstrated exact matches between FAERS and Recall records, with no observed formatting differences or entity resolution requirements. Unlike manufacturer names or other business-facing variables, these identifiers are not subject to product rebranding, naming variation, or organizational ownership changes.

Not all FDA-assigned identifiers demonstrated equal utility. UNII (Unique Ingredient Identifier) values are standardized identifiers used by the FDA to uniquely identify substances and ingredients regardless of naming conventions or synonyms. Although UNII values appear promising as substance-level identifiers, they **ARE NOT** consistently available across the datasets evaluated in this study. This finding highlights an important limitation of regulatory integration efforts: even highly standardized identifiers may not be uniformly available across all public FDA data sources.

EDARBI CASE STUDY

To better visualize potential linkage opportunities, a focused case study was conducted using the drug product EDARBI. The objective of the case study was to evaluate whether the candidate linkage variables identified during profiling could successfully connect records across Orange Book, FAERS, and Recall Enforcement Reports. Because application number demonstrated strong availability across all three

datasets and emerged as one of the most promising linkage variables during the preliminary integration assessment, it was selected as the primary retrieval mechanism for the investigation.

Orange Book and FAERS records were retrieved using application number “200796”, while Recall Enforcement Reports records were retrieved using “NDA200796”. The application numbers were identified through the profiling activities described previously and were used to retrieve records from each data source. The resulting linkage evidence is summarized in Table 6.

Identifier Group	Orange Book Value	FAERS Value	Recall Value	Requires Entity Resolution
Application Number	200796	200796	NDA200796	N
Brand Name	EDARBI	EDARBI	EDARBI	N
Active Ingredient	AZILSARTAN KAMEDOXOMIL	AZILSARTAN KAMEDOXOMIL	AZILSARTAN KAMEDOXOMIL	N
Manufacturer	AZURITY PHARMACEUTICALS INC	US-ARBOR PHARMACEUTICALS LLC-US-2015ARB000514	Azurity Pharmaceuticals Inc.	Y
Manufacturer Short Name	AZURITY	ARBOR PHARMACEUTICALS	AZURITY PHARMACEUTICALS INC.	Y
Product NDC		60631-040	60631-040	N
Package NDC		60631-040-07	60631-040-07	N
RXCUI		1091646	1091646	N
RXCUI		1091650	1091650	N
SPL ID		2f63b0e8-977a-4d22-9550-29ad1926a9de	2f63b0e8-977a-4d22-9550-29ad1926a9de	N
SPL Set ID		52b27c75-9f5a-4816-bafd-dace9d7d2063	52b27c75-9f5a-4816-bafd-dace9d7d2063	N

Table 6. EDARBI Cross-Dataset Linkage Evidence

Active ingredients also demonstrated strong linkage potential. Orange Book variable `name`, FAERS variable `activesubstancename`, and Recall Enforcement Reports variable `substance_name1` all contained the ingredient “AZILSARTAN KAMEDOXOMIL”. Because active ingredients represent chemical identity rather than marketing identity, they are less susceptible to rebranding activities and naming variation. Exact overlap was observed across all three datasets, making active ingredient one of the strongest human-readable linkage variables identified during the study. An interesting observation emerged during evaluation of active ingredients within FAERS. Although EDARBI was the product of interest, the resulting records frequently contained additional active ingredients associated with other medications taken by the same patient. This finding reflects the event-centric structure of FAERS and reinforces the observation that adverse event reports often contain information related to multiple products simultaneously.

Brand name represented another highly successful linkage variable. The value EDARBI appeared consistently across Orange Book, FAERS, and Recall Enforcement Reports without observed spelling variations or alternate naming conventions. Compared with active ingredients, brand names are immediately recognizable to a wider audience and can be easily validated through manual inspection. The consistency observed across all three datasets made brand name one of the most straightforward linkage variables evaluated during the investigation. Brand names are also more granular than active

ingredients, where typically multiple brand names may share the same active ingredient. Thus, active ingredients alone would not suffice for a linkage variable when trying to join tables.

Manufacturer information produced one of the most interesting findings of the entire case study. Unlike active ingredients and brand names, manufacturer values did not immediately align across datasets. Orange Book identified the manufacturer as “AZURITY PHARMACEUTICALS INC”, while Recall Enforcement Reports records contained a similar value. However, corresponding FAERS records referenced “ARBOR PHARMACEUTICALS” through manufacturer-related fields. At first glance, these records appeared inconsistent and potentially unsuitable for linkage.

Further investigation revealed that the apparent discrepancy reflected a real corporate event rather than a data-quality issue. In August 2021, Azurity Pharmaceuticals announced its acquisition of Arbor Pharmaceuticals, and the two organizations subsequently merged under the Azurity name. As a result, historical records associated with the same product reference either Arbor Pharmaceuticals or Azurity Pharmaceuticals. This discovery demonstrated that apparent identifier inconsistencies can sometimes reflect organizational history rather than errors in the underlying data. While application number, active ingredient, and brand name linked successfully with little additional effort, manufacturer information required external investigation and contextual interpretation before relationships could be established (Azurity Pharmaceuticals to Acquire and Merge with Arbor Pharmaceuticals 2021). This finding highlights the distinction between simple data transformation and true entity resolution. Standardization techniques can address differences in capitalization, punctuation, spacing, and formatting. In contrast, manufacturer records may require understanding organizational history, acquisitions, mergers, ownership changes, or evolving reporting relationships. As a result, manufacturer information was identified as one of the most challenging candidate linkage variables evaluated during the study.

FDA-assigned identifiers also performed exceptionally well during the case study. Application number demonstrated exceptionally strong linkage potential. Records were successfully identified across all three datasets, and the only transformation required was removal of the “NDA” prefix contained within Recall Enforcement Reports. Aside from this minor formatting difference, the values aligned exactly across sources. These findings suggest that application number represents one of the strongest candidate linkage variables identified during the investigation. However, additional products should be evaluated in future work because application numbers may not be consistently available across all products and datasets.

Product NDC, package NDC, RXCUI, SPL ID, and SPL set ID all demonstrated exact overlap between the datasets in which they were available. Unlike many human-readable variables, these identifiers required no entity resolution activities and exhibited little or no variability. The consistency observed among these identifiers reinforces their potential value in future automated integration efforts.

Collectively, the EDARBI case study demonstrated successful linkage through application number, active ingredient, brand name, product NDC, package NDC, RXCUI, SPL ID, and SPL set ID while simultaneously highlighting the limitations of manufacturer information as a standalone linkage variable. The investigation provides practical evidence that meaningful integration opportunities exist across publicly available FDA datasets while also illustrating the types of transformation and entity resolution challenges that must be addressed when constructing a broader lifecycle-oriented analytical resource.

CONCLUSION

KEY FINDINGS AND DISCUSSION

Public FDA datasets provide substantial opportunities for integration, but meaningful integration requires significantly more effort than simply acquiring the data. While OpenFDA has greatly simplified data retrieval, understanding how independently developed datasets relate to one another remains a substantial challenge. Differences in hierarchy, identifier availability, formatting conventions, and organizational context all create barriers that must be addressed before datasets can be reliably combined. Figure 2 summarizes the observed relationships between the three datasets, identifies the strongest candidate linkage variables, and highlights the major integration barriers discovered during the assessment.

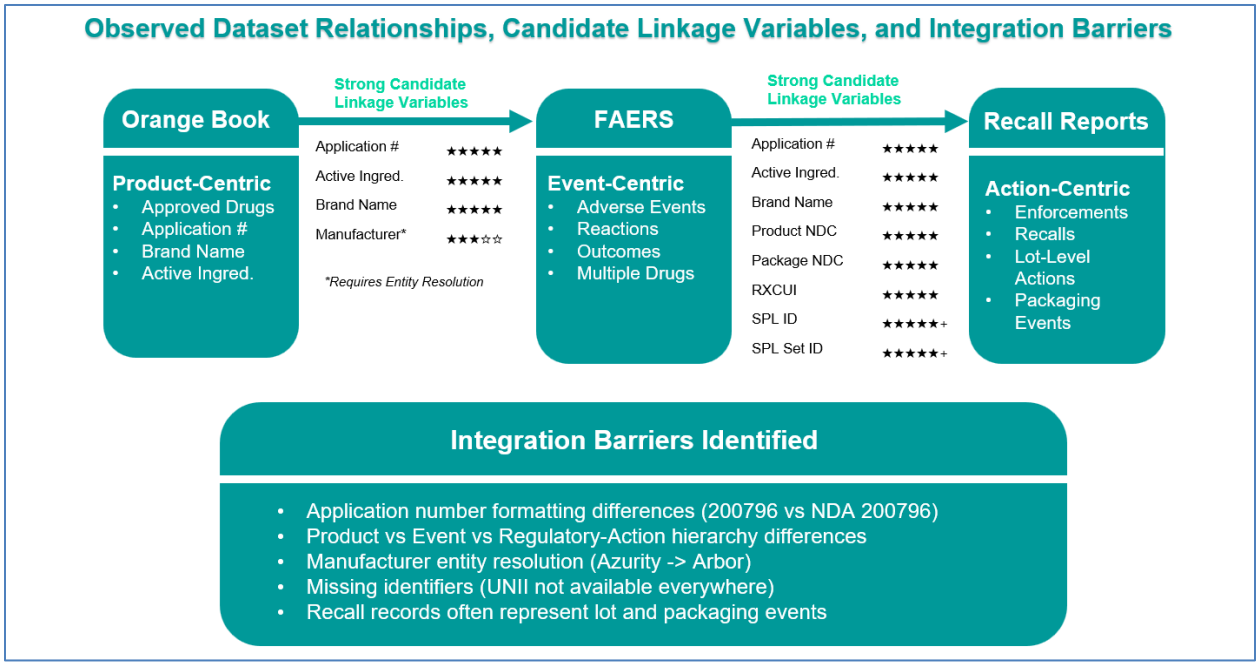


Figure 2. Observed Dataset Relationships, Candidate Linkage Variables, and Integration Barriers

One of the more surprising findings from this investigation was the relative performance of human-readable and FDA-assigned identifiers. At the outset of the study, it was expected that human-readable variables such as brand name, active ingredient, and manufacturer name would provide the most practical linkage opportunities because they are intuitive, easily interpreted, and straightforward to validate manually. In many cases, this assumption proved correct. Active ingredients and brand names demonstrated strong linkage potential and frequently appeared across multiple datasets with relatively little transformation required. Active ingredients proved particularly stable because they represent chemical identity rather than marketing identity, while brand names were generally easy to validate manually despite potential vulnerability to future naming variation and rebranding.

FDA-assigned identifiers frequently demonstrated even stronger integration characteristics. Application numbers demonstrate exceptionally strong linkage potential despite minor formatting differences such as “NDA” prefixes present in Recall data. Product NDC, package NDC, RXCUI, SPL ID, and SPL set ID often matched exactly across datasets (when available) and generally required no entity resolution activities. Unlike human-readable variables, these identifiers are designed specifically to represent products and product concepts in a standardized manner. As a result, they are less vulnerable to the naming inconsistencies, abbreviations, spelling variations, and organizational changes that affect business-facing variables. However, these variables were not observed in the Orange Book dataset and cannot be used alone to merge all three datasets.

Overall, the results suggest that neither identifier category should be used independently. Human-readable variables offered broader availability and easier validation, while FDA-assigned identifiers generally provided more deterministic linkage opportunities when available. The most promising integration strategy appears to involve a combination of both approaches, using FDA-assigned identifiers whenever available while supplementing them with human-readable variables to provide additional context, verification, and fallback linkage opportunities.

An unexpected finding emerged from the Recall Enforcement Reports. The original lifecycle hypothesis assumed that adverse event reports and recall actions would represent sequential stages within a common product narrative. However, many recall records were associated with manufacturing lots, packaging issues, labeling concerns, contamination events, or other operational factors rather than trends observed within adverse event reporting systems. This finding suggests that recall activity may not always serve as a direct endpoint within a simple product → adverse event → recall lifecycle model. This observation highlights an important distinction between post-market surveillance and regulatory

enforcement activities. While adverse event reporting focuses on health-related outcomes associated with product use, enforcement actions may originate from inspections, manufacturing deviations, quality testing, labeling discrepancies, or supply-chain issues. As a result, the relationship between adverse event reports and recall events appears more complex and less deterministic than originally anticipated.

The greatest challenge identified during the study involved entity resolution. Manufacturer information highlighted how organizational changes, acquisitions, and differing reporting structures can complicate cross-dataset integration efforts. The EDARBI case study demonstrated that records referring to the same product may contain different manufacturer values due to acquisitions, corporate restructuring, or historical reporting differences. These findings suggest that future lifecycle integration frameworks will likely require a combination of standardized identifiers, business rules, normalization procedures, and entity resolution methods rather than reliance on any single variable.

Although this study does not produce a fully integrated FDA lifecycle dataset, it demonstrates that meaningful linkage opportunities exist within publicly available regulatory data. More importantly, the study shows that the primary challenges associated with public FDA data integration are not related to data acquisition, but rather to understanding differences in hierarchy, identifier availability, formatting conventions, and entity resolution requirements. Based on the findings of this study, a future modeling-ready dataset could combine approved product metadata from Orange Book, adverse event information from FAERS, and regulatory action metadata from Recall Enforcement Reports. Application numbers, active ingredients, product NDCs, RXCUIs, SPL IDs, and SPL set IDs emerged as the strongest candidate linkage variables and would likely form the foundation of such an integrated resource. While additional work remains, particularly with respect to manufacturer normalization, entity resolution, and expansion to additional FDA datasets, the results suggest that lifecycle-oriented analysis using publicly available FDA data is both feasible and practical.

FUTURE WORK

This study focused on three publicly available FDA datasets and a single case-study product. Although meaningful linkage opportunities were identified, the broader FDA data ecosystem contains many additional data sources that could further enrich lifecycle-oriented analysis. Future work should expand the framework to include additional regulatory datasets such as medical device reports, vaccine adverse event records, clinical trial metadata, inspection data, warning letters, and other FDA regulatory systems. Incorporating these resources would provide a more comprehensive view of the broader medical product lifecycle and potentially reveal additional linkage opportunities not observed within the current study.

A second area for future investigation involves automated entity resolution. The EDARBI case study demonstrated that organizational names may change over time due to acquisitions, mergers, corporate restructuring, and evolving reporting practices. Traditional approaches such as standardization, deterministic matching, and fuzzy matching remain promising techniques for resolving these discrepancies. In addition, emerging LLM-assisted approaches may provide new opportunities for reconciling organizational names, product descriptions, and other semi-structured regulatory information across large datasets. Further research is needed to evaluate the accuracy, scalability, and practical applicability of these methods within FDA data integration workflows.

The findings of this study also provide a foundation for development of a broader modeling-ready analytical resource. A future integrated dataset could combine approved product information from Orange Book, adverse event information from FAERS, and regulatory action metadata from Recall Enforcement Reports into a unified lifecycle-oriented structure. Application numbers, active ingredients, product NDCs, RXCUIs, SPL IDs, and SPL set IDs emerged as particularly strong candidate linkage variables and would likely form the backbone of such a resource. While substantial work remains, particularly with respect to hierarchy reconciliation and entity resolution, the results suggest that construction of a publicly available lifecycle-oriented analytical dataset is both feasible and practical.

REFERENCES

Azurity Pharmaceuticals. 2021. “Azurity Pharmaceuticals to Acquire and Merge with Arbor Pharmaceuticals.” Available at: https://www.azurity.com/news/acquisition-of-arbor-pharmaceuticals. Accessed September 2026.

Congressional Research Service. 2022. *The OPEN Government Data Act: A Primer*. Available at: https://www.congress.gov/crs_external_products/IF/PDF/IF12299/IF12299.4.pdf. Accessed September 2026.

Giunchi, V., M. Fusaroli, M. Hauben, E. Raschi, and E. Poluzzi. 2023. “Challenges and Opportunities in Accessing and Analysing FAERS Data: A Call Towards a Collaborative Approach.” *Drug Safety*, 46(10): 921-926.

National Library of Medicine. 2024. “RxNorm.” Available at: https://www.nlm.nih.gov/research/umls/rxnorm/. Accessed September 2026.

U.S. Food and Drug Administration. “FDA Adverse Event Reporting System (FAERS).” *openFDA*. Available at: https://open.fda.gov/data/faers/. Accessed September 2026.

U.S. Food and Drug Administration. “Approved Drug Products with Therapeutic Equivalence Evaluations (Orange Book).” *FDA Drug Approvals and Databases*. Available at: https://www.fda.gov/drugs/drug-approvals-and-databases/approved-drug-products-therapeutic-equivalence-evaluations-orange-book. Accessed September 2026.

U.S. Food and Drug Administration. “Drug Enforcement Reports.” *openFDA*. Available at: https://open.fda.gov/apis/drug/enforcement/. Accessed September 2026.

U.S. Food and Drug Administration. “Structured Product Labeling Resources.” Available at: https://www.fda.gov/industry/fda-data-standards-advisory-board/structured-product-labeling-resources. Accessed September 2026.

CONTACT INFORMATION

Your comments and questions are valued and encouraged. Contact the author at:

Shelly Hunt
Siemens Digital Industry Software
+1 984-400-2949
Shelly.Hunt@Siemens.com